

Building Next-Generation Government Applications *Using* Open Source Technologies

Governments the world over are trying to provide citizen-centric services. Our own country has taken the 'Smart City' initiative. The idea is to provide better governance, which can be achieved by the digitisation of services. Open source software can bring such services to the citizen accurately, speedily, securely and in a user friendly manner.



The success of digital governance depends on the ability to enable multiple government agencies and departments to communicate with each other. All the work that has been done earlier should be interoperable with all the new initiatives to be implemented in the near future. In addition, the focus is on enabling the best citizen experience, making available the relevant information and being responsive by leveraging the latest technology trends (like analytics). This is the key to digital governance, as it ensures smooth communication between departments to make the citizen the centre of all service delivery, bringing about better governance and higher accountability.

The adoption of open source software (OSS) in the government sector has increased worldwide and has led to innovations in the implementation of ICT solutions across the public and private domains. The use of OSS in the key domains of ICT implementation (like application development, Internet connectivity, infrastructure, data centres and the mobile) has helped in the widespread adoption of open source technologies by various customers across the world.

In India, the Centre has notified the guidelines for adopting open source software by government organisations.

These organisations are encouraged to adopt OSS in all e-governance systems rather than closed source software (CSS). In addition, the Indian government has made OSS mandatory for all new e-governance applications as well as new versions of legacy and existing systems.

Many governments across the globe like Australia, the states of British Columbia and Quebec in Canada, France, India, Italy, the United Kingdom and the United States have published OSS policies that cover information and communications technology procurement processes, the adoption of OSS solutions, etc.

Fortune 500 businesses are using OSS to build enterprise applications. According to Gartner and Forrester, open source has acquired a greater share of the global enterprise software market while proprietary software adoption is reducing significantly.

The cloud, Big Data, content management systems, responsive Web design, mobility, databases, operating systems, infrastructure components, enterprise integration, development tools and analytics are the major OSS technologies being adopted successfully in the e-governance space.

In this context, let us explore the high-level architecture and related infrastructure of a comprehensive software platform